

MECB2101T NUMERICAL METHODS AND APPLICATIONS

Lectures: 3, Tutorials: 1, Practicals: 0			Credits: 4.0
Internal Evaluation:	Maximum Marks: 50	Minimum Passing Marks: 40%	
External Exam.:	Maximum Marks: 50	Minimum Passing Marks: 40%	Time Allowed: 3 Hours

Course Objectives: The course aims to: 1. Provide understanding of error analysis and numerical methods for solving algebraic and transcendental equations; 2. Develop skills to solve systems of linear equations using direct and iterative techniques; 3. Introduce methods for computing eigenvalues, eigenvectors, and curve fitting using least squares; 4. Apply numerical techniques for differentiation and integration; 5. Enable solution of ordinary differential equations using standard numerical methods.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Apply numerical methods with error estimation to solve algebraic and transcendental equations; 2. Solve systems of linear equations and compute eigenvalues and eigenvectors using appropriate techniques; 3. Perform curve fitting using least squares methods; 4. Apply numerical differentiation and integration techniques; 5. Solve initial and boundary value problems of differential equations using numerical methods.

SECTION-A

Solution of Algebraic and Transcendental Equations: Truncation error; Round-Off error; Absolute and Relative errors; Bisection method; Iteration method: Conditions for the convergence of the Iteration method, Acceleration of convergence: Aitken's Δ^2 process; Newton-Raphson method: Conditions for the convergence; Comparison of Regula Falsi method and Secant method; Rate of convergence and geometrical representation of each method; Newton-Raphson method for system of non-linear equations.

Solution of system of non-homogeneous linear equations: Matrix inversion method; Gauss Elimination method: Partial and Complete Pivoting.; Gauss Jordan Elimination method; Triangularization method; Factorization method; Jacobi's method and Gauss-Seidal's method. Solution of Tridiagonal system of equations.

Eigen values and Eigen vectors of a matrix: Eigen values of Transpose of matrix, inverse of matrix, Hermitian matrix and Similar matrices Diagonalization of a matrix; Iterative Methods to find Eigen values and Eigen vectors: Power method and Jacobi method.

Curve fitting: Method of Least Squares-fitting a straight line, a second-degree parabola and exponential curves.

SECTION-B

Numerical Differentiation and Integration: Numerical Differentiation using finite differences, Numerical Integration; Newton-Cotes methods, Trapezoidal rule, Simpson's 1/3 rule, Simpson's 3/8 rule.

Numerical Solution of Differential Equations: Numerical solution of first order ordinary differential equations using Taylor's series method; Picard's method; Euler's method; Improved Euler's method; Modified Euler's method; Runge-Kutta method of Second and Fourth order; Predictor-Corrector methods: Milne's method and Adam's method. Boundary values problems for ordinary differential equation by finite difference method.

Recommended Books:

1. Numerical Methods for Mathematics, Science and Engineering by John H. Mathews (Prentice-Hall of India)
2. Numerical Methods for Scientist and Engineering Computation by M. K. Jain, S. R. K. Iyengar and R. K. Jain (New Age International Publisher)
3. Introductory Methods of Numerical Analysis by S. S. Sastry (PHI Learning)

MECB2102T**OPERATIONS RESEARCH**

Lectures: 3, Tutorials: 0, Practicals: 0			Credits: 3.0
Internal Evaluation:	Maximum Marks: 50	Minimum Passing Marks: 40%	
External Exam.:	Maximum Marks: 50	Minimum Passing Marks: 40%	Time Allowed: 3 Hours

Course Objectives: The course aims to: 1. Develop understanding of the fundamental concepts, scope, and applications of Operations Research; 2. Introduce mathematical modelling techniques for decision-making problems; 3. Provide knowledge of optimization techniques such as Linear Programming, Transportation, and Assignment models; 4. Familiarize students with decision-making tools including Game Theory, PERT/CPM, and Simulation; 5. Enable analytical thinking for solving complex engineering and management problems.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Formulate and analyze real-world problems using Operations Research techniques; 2. Solve Linear Programming problems using graphical and simplex methods including duality and sensitivity analysis; 3. Solve Transportation, Assignment, and related optimization problems; 4. Apply Game Theory concepts for decision-making under competitive situations; 5. Analyze project scheduling problems using PERT/CPM and apply simulation techniques for problem-solving.

SECTION-A

Introduction to operations Research: Definitions, characteristics, necessity, scope and objectives of O.R. Phases of Operations Research study, Limitations of O.R.

Linear Programming: Assumptions in L.P. formulation of mathematical models for various types of L.P. problems, graphical method of solving L.P. problems, Simplex method, artificial variable technique-the big-M method, The two-phase method, degeneracy in simplex method, Unconstrained variables, duality in Linear Programming, Sensitivity analysis, Limitations of L.P. models. Further developments of Linear Programming

Transportation & Assignment Model: Formulation and solution of transportation problems, Assignment model, formulation and solution of assignment problems, The travelling salesman problem.

SECTION-B

Game Theory: Decision making environments, Decision trees, rules and solution of games with and without saddle point, rules of Dominance, arithmetic and algebraic methods for 2x2 games.

PERT & CPM Models: Characteristics & uses, drawing of network, removal of redundancy in network, Forward and Backward pass computations, Free Slack, Total slack, float, Critical path, Crashing.

Simulation: Advantages and limitations of simulation, Monte Carlo method, Generation of random numbers, solution of problems using simulation.

MECB2103T BASIC THERMODYNAMICS

Lectures: 3, Tutorials: 0, Practicals: 0			Credits: 3.0
Internal Evaluation:	Maximum Marks: 50	Minimum Passing Marks: 40%	
External Exam.:	Maximum Marks: 50	Minimum Passing Marks: 40%	
			Time Allowed: 3 Hours

Course Objectives: The course aims to: 1. Develop understanding of fundamental concepts of thermodynamics, systems, and energy interactions; 2. Introduce the First and Second Laws of Thermodynamics and their engineering applications; 3. Provide knowledge of thermodynamic processes in closed and open systems; 4. Familiarize students with entropy, thermodynamic cycles, and performance analysis of systems; 5. Develop understanding of steam properties and air standard cycles.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Explain basic thermodynamic concepts, systems, and properties of matter; 2. Apply the First Law of Thermodynamics to closed and open systems; 3. Analyze thermodynamic processes such as isothermal, isobaric, isochoric, adiabatic, and polytropic processes; 4. Apply the Second Law of Thermodynamics and entropy concepts to engineering systems; 5. Analyze air standard cycles and steam properties for performance evaluation of thermal systems.

Section-A

Basic Concepts of Thermodynamics: Thermodynamics system, boundary and surroundings. Open, closed and isolated systems. State, property, process and cycle. Reversible, quasi-static and irreversible processes and conditions for reversibility. Energy and its forms; energy transfer across system boundaries, heat and work, property and energy as point and path functions, Ideal gas and characteristic gas equation. Zeroth law of thermodynamics, Concept of thermal equilibrium and principle of thermometry.

First Law of Thermodynamics: Essence and corollaries of the first law, analytical expressions applicable to a process and cycle; internal energy, enthalpy and specific heats; first law analysis of steady flow, applications of steady flow energy equation to engineering devices.

Applications of First Law of Thermodynamics: Closed and open systems, analysis of non-flow and flow processes for an Ideal gas under constant volume (Isochoric), constant pressure (Isobaric), constant temperature (isothermal), adiabatic and polytropic conditions. Analysis of free expansion and throttling processes, Representation of these processes on P-V charts and analysis of property changes and energy exchange (work and heat) during these processes.

Section-B

Second law of Thermodynamics: Limitations of first law, various statements of second law and their equivalence, application of statements of second law to heat engine, heat pump and refrigerator, Philosophy of Carnot cycle and its consequences. Carnot theorem for heat engines & heat pump, Clausius inequality, concept and philosophy of entropy and entropy changes during various processes, Temperature – entropy chart and representation of various processes on it, Third law of thermodynamics.

Thermodynamic Cycles: Air standard cycles and air standard efficiency, index of performance and mean effective pressure, Otto cycle, Diesel Cycle and Dual cycle, comparison of Otto, diesel and dual cycles.

Properties of Steam: Pure substance constant pressure formation of steam, steam tables, constant volume, constant pressure and isentropic processes.

MECB2104T STRENGTH OF MATERIALS

Lectures: 3, Tutorials: 1, Practicals: 0			Credits: 4.0
Internal Evaluation:	Maximum Marks: 50	Minimum Passing Marks: 40%	
External Exam.:	Maximum Marks: 50	Minimum Passing Marks: 40%	Time Allowed: 3 Hours

Course Objectives: The course aims to: 1. Develop understanding of fundamental concepts of stress, strain, and mechanical properties of materials; 2. Introduce analysis of axially loaded members, beams, shafts, and columns under different loading conditions; 3. Provide knowledge of shear force, bending moment, bending stresses, and torsion in structural members; 4. Familiarize students with deflection of beams, stability of columns, and stress transformation; 5. Introduce theories of failure and their applications in design of structural elements.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Analyze stress-strain behavior and mechanical properties of materials; 2. Solve problems on axially loaded members including thermal stresses; 3. Draw shear force and bending moment diagrams for various beam configurations; 4. Determine bending, shear, torsional stresses, and deflection in structural members; 5. Apply theories of failure, Mohr's circle, and stress transformation in engineering design.

Section-A

Basic Concepts: Free body diagram, Types of stresses and strains, Stress-strain curve for Ductile & Brittle materials, Types of Moduli for materials, Poisson's ratio.

Axially Loaded Bars: Analysis of axially loaded bars (Statically determinate and indeterminate problems), Bars in series and parallel, Thermal stresses.

Shear Force and Bending Moment Diagrams: Types of- loads, supports and of beams, SFD & BMD of- simply supported, over hung and cantilever beams (subjected to point loads, uniformly distributed loads and moments).

Bending and Shear Stresses in Beams: Theory of pure bending, Flexural formula (derivation), Bending stresses in beams of various cross-section, Shear stresses in beams (derivation), Distribution of shear stresses in beams of various cross-section.

Torsion of Circular Shafts: Torsional stresses in circular shafts subjected to torsional moment (torque), Torsion formula (derivation), Power transmitted by shafts, Design of shafts based on strength and torsional stiffness, Comparison of solid and hollow shafts.

Section-B

Deflection of Beams: Governing differential equation for deflection of straight beams having constant flexural rigidity (no proof), Determination of slope and deflection for different types of beams (Double integration and Macaulay's methods).

Stability of Columns: Buckling of columns under different end conditions (Euler's Theory), Limitations of Euler's Formula, Rankine's Formula.

Compound Stress and Strain: Generalized Hooke's law, 2D state of stress (Plane stress condition), Analysis of 2D stress element (Stress transformation equations-**Derivation**), Principal planes, Principal normal and shear stresses, Mohr's stress circle, Strain transformation equations (**no derivation**), Principal normal and shear strains, Mohr's strain circle, Deriving principal stresses from principal strains, Strain rosette.

Theories of Failure: Major theories of elastic failure (**no derivations**), Graphical comparison of theories of failure, Application of theories of failure in design of members subjected combined bending and torsion.

MECB2105T MANUFACTURING TECHNOLOGY

Lectures: 3, Tutorials: 0, Practicals: 0			Credits: 3.0
Internal Evaluation:	Maximum Marks: 50	Minimum Passing Marks: 40%	
External Exam.:	Maximum Marks: 50	Minimum Passing Marks: 40%	Time Allowed: 3 Hours

Course Objectives: The course aims to: 1. Develop understanding of metal casting processes, moulding sand properties, and testing methods; 2. Introduce machine moulding techniques and special casting processes; 3. Provide knowledge of powder metallurgy, welding processes, and weldability concepts; 4. Familiarize students with machining processes including milling, drilling, grinding, and abrasive finishing; 5. Introduce metal forming processes and basic plastics processing techniques.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Analyze properties and testing of moulding sand used in metal casting; 2. Explain machine moulding and various special casting processes; 3. Understand powder metallurgy and welding processes including defects and applications; 4. Apply knowledge of machining processes such as milling, drilling, and grinding; 5. Describe metal forming processes and plastics manufacturing techniques in engineering applications.

Section-A

Metal Casting: Sand testing, significance of sand testing, determination of grain fineness number, clay content, permeability number, shatter index, compactability, green strength and dry strength of moulding sand; sand behaviour with changing moisture content, clay content and degree of ramming.

Machine Moulding: Types of moulding machines, Squeeze, Jolt, Jolt-Squeeze and sand slinger moulding machines.

Special Casting Processes: Shell moulding, Investment casting, Die casting, Hot chamber and Cold chamber die casting processes, Centrifugal casting, Inspection of casting, Casting defects, their causes and mitigation.

Powder Metallurgy (PM): Characteristics, applications, comparison with other manufacturing processes, steps involved in powder metallurgy, methods of powder preparation, primary and secondary processes involved in powder metallurgy.

Welding: Welding and its classification, Weldability, Factors affecting weldability, Principles and application of TIG and MIG welding, Friction and Inertia welding, Submerged arc welding, Welding defects- types, causes and prevention.

Section-B

Metal Machining: Milling, up & down milling processes, their applications and limitations, types of milling cutters, selection of milling cutter for different operations, milling machines, Knee & Column type milling machines, Bed type milling machines, Planer type milling machines, Universal milling machines, Introduction to indexing, Types of indexing (detailed description of simple indexing only).

Drilling: Twist drill geometry, drilling processes, drilling machines.

Grinding: Types of abrasives and bonds, types of grinding wheels, selection of grinding wheels, Grinding processes- Cylindrical grinding, Surface grinding, Centreless grinding.

Other abrasive finishing processes: Lapping, honing, and buffing process.

Metal Forming: High energy rate forming processes- Explosive forming, Electro-hydraulic forming, Electro-magnetic forming, applications of metal forming processes.

Plastics & its Processing: Plastics, differences between thermosetting and thermoplastics plastics, applications of thermosetting and thermoplastics plastics, basic manufacturing processes for processing of plastics.

MECB2106P MACHINE DRAWING

Lectures: 1, Tutorials: 0, Practicals: 4		Credits: 3.0
Evaluation: Internal only	Maximum Marks: 100	Minimum Passing Marks: 40%

Course Objectives: The course aims to: 1. Develop understanding of engineering drawing conventions, symbols, and standards; 2. Introduce representation of fasteners, riveted joints, pulleys, and pipe joints; 3. Provide knowledge of keys, couplings, bearings, and brackets used in machine design; 4. Familiarize students with standard components such as screw jacks, piston and connecting rods; 5. Enable interpretation and application of machine elements in engineering drawings.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Interpret standard engineering drawing symbols and tolerances; 2. Draw and identify fasteners, rivets, and pulley systems; 3. Represent pipe joints, keys, and shaft couplings in engineering drawings; 4. Identify and sketch different types of bearings and brackets; 5. Understand and apply knowledge of machine elements in design drawings.

Section-A

Introduction: Symbols of standard tolerances, machining symbols, Surface finish and welding symbols, sectioning representation, dimensioning principals, Familiarization of Code IS: 296.

Fastenings: Forms of screw threads, conventional representation of single and multiple start threads, left-hand and right-hand threads, sketching of square and hexagonal nuts, types of nuts, sketching of square headed and hexagonal headed bolts, types of bolts, studs, sets screws, types of locking arrangements.

Rivets and Riveted Joints: Shapes of rivet heads, lap and butt joints, single and multiple riveted joints, straight and zig-zag riveting.

Knuckle joints & Pulleys: V-belt pulley, Flat belt pulley.

Section-B

Pipe Joints: Flange joints, spigot-socket & expansion joints, Union joints.

Keys and Shaft Couplings: Types of keys, comparison between keys and cotters. Couplings, classifications, Rigid flange couplings and muff couplings, flexible pin type coupling, Oldham's coupling, Claw Coupling, Universal Coupling. Knuckle and Cotter joints.

Bearings and Brackets: Solid journal bearing, bushed bearing, Pedestal bearing, footstep bearing, swivel bearings. Wall brackets.

Miscellaneous: Screw Jack, Tail Stock, Piston and connecting rods.

ERSB2021Q Environmental and Road Safety Awareness

Session: 2025-26, 2026-27, & 2027-28

Lectures: 2, Tutorials: 0, Practicals: 0			Credits: 0 (Qualifying Only)
Internal Evaluation:	Maximum Marks: 30	Minimum Passing Marks: 40%	
External Exam.:	Maximum Marks: 70	Minimum Passing Marks: 40%	Time Allowed: 3 Hours

INSTRUCTIONS FOR THE PAPER SETTERS (Regular Students)

The question paper will consist of three sections A, B and C. Each of sections A and B will have four questions from the respective sections of the syllabus. Each question shall carry 10 marks. Section C will consist of 10 short answer type questions of 3 marks each.

INSTRUCTIONS FOR THE CANDIDATES

Candidates are required to attempt any two questions from each section A and B. Section C is compulsory.

SECTION-A**INTRODUCTION TO ENVIRONMENTAL STUDIES:**

The multidisciplinary nature of environmental studies. Definition, scope and importance

Concept of Biosphere – Lithosphere, Hydrosphere, Atmosphere.

ECOSYSTEM & BIODIVERSITY CONSERVATION:

Ecosystem and its components, Types of Ecosystems

Biodiversity - Definition and Value, Threats to biodiversity and its conservation

Level of biological diversity: genetic, species and ecosystem diversity; bio-geographic zones of India; biodiversity patterns and global biodiversity hot spots.

India as Mega-biodiversity nation; Endangered and endemic species of India.

Ecosystem and biodiversity services: Ecological, economic, social, ethical, aesthetic and informational value.

NATURAL RESOURCES-RENEWABLE AND NON-RENEWABLE RESOURCES:

Land resources and land use change; land degradation, soil erosion and desertification.

Deforestation: causes and impacts due to mining, dam building on environment, Forests, Biodiversity and tribal populations.

Water: Use and over-exploitation of surface and ground water, Floods, droughts, conflicts over water (international & inter-state)

Energy resources: renewable and nonrenewable energy sources, use of alternate energy sources, growing energy needs, case studies.

ENVIRONMENTAL POLLUTION

Environmental Pollution: types, causes, effects and controls; Air, Water, Soil and noise pollution. Nuclear hazards and human health risks Solid waste management, Source Segregations: Control measures of urban and Industrial waste. Pollution case studies.

SECTION-B**ENVIRONMENTAL PROTECTION LAWS IN INDIA:**

Environmental protection act for; Air (Prevention and control of pollution), Water (Prevention and Control of pollution), Wild life, Forest Conservation, Issues involved in the enforcement of environmental legislation; Role of an individual in prevention of pollution; Environmental policies & Practices; Climate change, global warming, ozone layer depletion, acid rain and impacts on human communities and agriculture.

HUMAN COMMUNITIES AND THE ENVIRONMENT:

Human population growth: Impacts on environment, human health and welfare, Sanitation & Hygiene. Resettlement and rehabilitation of project affected persons; case studies. Disaster management: floods, earthquake, cyclones and landslides.

Environment movements: Chipko, Silent valley, Bishnois of Rajasthan. Environmental ethics: Role of Indian and other religions and cultures in environmental conservation for a Clean-green pollution free state.

Environmental communication and public awareness, case studies (e.g., CNG vehicles in Delhi)

ROAD SAFETY AWARENESS:

Concept and significance of Road safety, Traffic signs, Traffic rules, Traffic Offences and penalties, How to obtain license, Role of first aid in Road Safety.

Stubble Burning; Meaning of Stubble burning.; Impact on health & environment; Management and alternative uses of crop stubble.

Environmental Legislations and Policies for Restriction of Agriculture Residue Burning in Punjab.

FIELD WORK:

Visit to an area to document environmental assets: river/Forest/Flora/Fauna, etc.

Visit to Local polluted site –Urban/Rural/Industrial/Agricultural.

Study of common Plants, Insects, Birds and basic principles of identification.

Study of simple ecosystems-pond, river, Delhi Ridge, etc.

DMPB2102Q DISASTER MANAGEMENT AND PREPAREDNESS (Qualifying Subject)

Lectures: 2, Tutorials: 0, Practicals: 0			Credits: 0 (Qualifying Only)
Internal Evaluation:	Maximum Marks: 50	Minimum Passing Marks: 40%	
External Exam.:	Maximum Marks: 50	Minimum Passing Marks: 40%	Time Allowed: 3 Hours

Course Objectives: The course aims to: 1. Develop understanding of basic concepts of disaster, hazard, vulnerability, risk, and capacity; 2. Introduce different types of disasters and their causes, impacts, and control measures; 3. Provide knowledge of disaster management cycle including preparedness, mitigation, response, and recovery; 4. Familiarize students with disaster management framework, policies, and institutional mechanisms in India; 5. Introduce applications of science and technology in disaster management.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Explain fundamental concepts of disaster and risk management; 2. Identify types, causes, and impacts of various natural and man-made disasters; 3. Apply knowledge of disaster management cycle for effective planning and response; 4. Understand disaster management policies, frameworks, and institutional structures in India; 5. Describe the role of science and technology in disaster mitigation and management.

SECTION-A

Understanding Disasters: Understanding the Concepts and definitions of Disaster, Hazard, Vulnerability, Risk, Capacity– Disaster and Development, and disaster management.

Types, Trends, Causes, Consequences and Control of Disasters: Geological Disasters (earthquakes, landslides, tsunami, mining); Hydro-Meteorological Disasters (floods, cyclones, lightning, thunderstorms, hail storms, avalanches, droughts, cold and heat waves); Biological Disasters (epidemics, pest attacks, forest fire); Technological Disasters (chemical, industrial, radiological, nuclear) and Man-made Disasters (building collapse, rural and urban fire, road and rail accidents, nuclear, radiological, chemicals and biological disasters); Global Disaster Trends; Emerging Risks of Disasters; Climate Change and Urban Disasters.

Disaster Management Cycle and Framework: Disaster Management Cycle– Paradigm Shift in Disaster Management; Pre-Disaster– Risk Assessment and Analysis, Risk Mapping, zonation and Microzonation, Prevention and Mitigation of Disasters, Early Warning System; Preparedness, Capacity Development; Awareness; During Disaster– Evacuation, Disaster Communication, Search and Rescue, Emergency Operation Centre, Incident Command System, Relief and Rehabilitation; Post-disaster– Damage and Needs Assessment, Restoration of Critical Infrastructure– Early Recovery, Reconstruction and Redevelopment; IDNDR, Yokohama Strategy, Hyogo Framework of Action.

SECTION-B

Disaster Management in India: Disaster Profile of India– Mega Disasters of India and Lessons Learnt; Disaster Management Act 2005– Institutional and Financial Mechanism National Policy on Disaster Management, National Guidelines and Plans on Disaster Management; Role of Government (local, state and national), Non-Government and Inter- Governmental Agencies.

Applications of Science and Technology for Disaster Management: Geo-informatics in Disaster Management (RS, GIS, GPS and RS); Disaster Communication System (Early Warning and Its Dissemination); Land Use Planning and Development Regulations; Disaster Safe Designs and Constructions; Structural and Non-Structural Mitigation of Disasters S&T Institutions for Disaster Management in India.

MECB2108P**STRENGTH OF MATERIALS LAB**

Lectures: 0, Tutorials: 0, Practicals: 2		Credits: 1.0
Evaluation: Internal only	Maximum Marks: 100	Minimum Passing Marks: 40%

Course Objectives: Course Objectives: The course aims to: 1. Provide hands-on experience with mechanical testing methods for evaluating material properties; 2. Develop skills in performing hardness, tensile, impact, and torsion tests; 3. Understand stress-strain behavior and mechanical characteristics of materials using standard testing machines; 4. Familiarize students with the operation of universal and specialized testing equipment.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Perform hardness tests such as Rockwell, Brinell, and Vickers methods; 2. Conduct tensile, torsion, and impact tests to determine mechanical properties; 3. Analyze stress-strain behavior and evaluate material parameters such as Young's modulus and impact strength; 4. Operate universal testing machines and interpret experimental results for material characterization.

List of Experiments:

1. To determine Rockwell hardness number of the specimen of steel / soft metal
2. To determine Brinell hardness number of the specimen of steel / soft metal
3. To determine Vicker's hardness number of the specimen of steel / soft metal
4. To determine the modulus of rigidity of a bar on torsion testing machine (destructive test)
5. To determine the impact strength of a specimen on Izod / Charpy impact testing machine
6. To determine the Young's modulus of the material of a beam simply supported at the ends and carrying a concentrated load at the center
7. To perform tensile test on a dumbbell shapes specimen (strip) on tensile testing machine and draw stress-strain curve to determine Young's modulus of the test material.
8. To study the behavior of the material on Universal Testing Machine

MECB2109P MANUFACTURING TECHNOLOGY LAB

Lectures: 0, Tutorials: 0, Practicals: 2		Credits: 1.0
Evaluation: Internal only	Maximum Marks: 100	Minimum Passing Marks: 40%

Course Objectives: The course aims to: 1. Provide hands-on training in sand testing, mould preparation, and basic foundry practices; 2. Develop practical skills in machining operations such as milling, gear cutting, drilling, and shaping; 3. Familiarize students with arc, MIG, and resistance welding processes and joint testing; 4. To understand common manufacturing defects and their prevention in casting and welding.

Course Outcomes (COs): After completing the laboratory, students will be able to: 1. Perform sand testing and prepare moulds for basic casting operations; 2. Execute fundamental machining operations and understand machine tool working principles; 3. Carry out basic welding operations and evaluate the strength and quality of welded joints; 4. Identify common casting and welding defects and suggest appropriate remedies.

List of Experiments:**Sand Testing Shop:**

1. Determination of the clay content in a sample of moulding sand.
2. Estimation of moisture content of the sample of green sand using Rapid Moisture teller / Rapid Drier.
3. To determine shatter index of the moulding sand.
4. To determine the permeability of a given sample of green sand and dried sand.
5. Determination of mould / core hardness using portable and sand hardness tester.
6. To prepare the green sand for sand moulding, Study of mechanical sand rammer for sand moulding. Relationship between mulling time & green strength, percentage clay contains and green compressive strength.

Foundry Shop:

7. Metal casting exercises (Pulley using Cheek, Hollow casting)
8. Study of casting defects and remedies.

Machine Shop:

9. Spur Gear cutting on Milling Machine.
10. Demonstration on milling machine: Up & Down milling.
11. Demonstration on Shaper, Drilling machines.

Welding Shop:

12. Demonstration on arc welding processes.
13. Testing the strength of a welded joint using MIG welding.
14. Demonstrations of various Resistance Welding Techniques.

Handwritten signatures and initials:
 AS
 J. K. S.
 A.